

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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Anonymous submission

ABSTRACT

Vision-language models (VLMs) now perform strongly on visual question answering, document understanding, multimodal dialogue, and other broad multimodal benchmarks. This paper highlights a complementary contribution: a focused evaluation of whether those gains translate to fine-grained visual knowledge. We evaluate 15 recent VLMs on established fine-grained classification benchmarks and compare these results with performance on general VLM benchmarks. The evaluation shows that fine-grained recognition is a distinct and under-measured axis of VLM capability, with large differences among models that otherwise have similar general benchmark scores and with persistent gaps between many VLMs and their underlying CLIP-style vision encoders. To explain this gap, we contribute 22 controlled ablations over model components and training choices. These ablations show that stronger LLMs improve fine-grained and general benchmark scores similarly, whereas stronger vision encoders disproportionately benefit fine-grained classification. We further find that pretraining is important for fine-grained performance, especially when language model weights are unfrozen during pretraining, while data-quality changes have more limited effects in our setting. Together, these results make fine-grained classification a clearer target for VLM evaluation and identify the design choices most associated with stronger vision-centric capability.

1 INTRODUCTION

Vision-language models (VLMs) combine vision encoders with large language models (LLMs) and have rapidly improved on tasks that require interpreting, describing, and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Their progress is especially visible on open-ended visual question answering, document understanding, and multimodal dialogue benchmarks (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). This paper foregrounds a capability that is less visible in those evaluations: fine-grained visual perception. We ask two reviewer-facing questions: how well do current VLMs recognize fine-grained categories, and which model or training choices most directly improve that ability?

Fine-grained visual recognition—the ability to distinguish between visually similar categories—requires attention to subtle distinguishing features while remaining robust to intra-class variation in pose, lighting, and background. Traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) are routinely evaluated on such benchmarks and often perform strongly. By contrast, newer LLM-based VLMs are usually assessed through VQA-style benchmarks, so their fine-grained recognition ability remains comparatively under-specified.

This missing measurement matters because fine-grained perception is a prerequisite for many downstream uses of VLMs. Real applications often involve long-tailed categories where adjacent labels have different consequences. For example, a model that cannot identify a mushroom species cannot reliably use textual knowledge to determine whether it is poisonous or safe to eat. More broadly, understanding which factors control fine-grained knowledge acquisition can inform architectures and training recipes for VLMs that need stronger vision-centric behavior, not only stronger language-mediated reasoning.

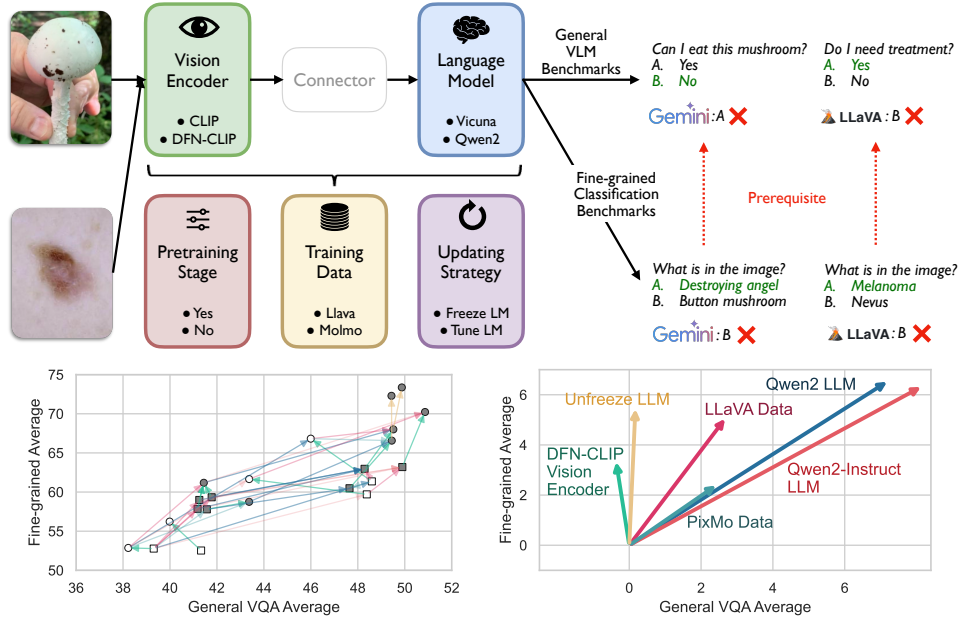


Figure 1: **Contribution overview.** (Top) We make fine-grained classification a visible evaluation target for VLMs, emphasizing a vision-centric capability that is often hidden by aggregate VQA-style benchmarks. (Bottom) We then use 22 controlled ablations over model components and training strategies (left) to identify which choices most strongly drive fine-grained classification performance (right).

Our first contribution is a broad evaluation that separates fine-grained visual knowledge from general VLM performance. We evaluate 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024), on four fine-grained classification benchmarks: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014). We compare these results with aggregated scores from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). This evaluation shows substantial variation in fine-grained accuracy among models with similar general abilities, indicating that fine-grained recognition is a distinct axis of visual intelligence that current VLM benchmarks do not fully capture. It also shows a persistent gap between many VLMs and their CLIP-based vision encoders, consistent with prior observations (Zhang et al., 2024; Geigle et al., 2024).

Our second contribution is a controlled ablation study that explains which design choices contribute to fine-grained classification performance. We test whether base components such as the language model and vision encoder, and training choices such as pretraining, weight updating, and data quality, affect fine-grained classification differently from general VQA performance. Across 22 ablations, we identify the following supported conclusions:

- **Language model (§4.2):** A stronger base language model improves both fine-grained classification and general VLM benchmarks, making it a broad performance contributor rather than a fine-grained-specific one.
- **Vision encoder (§4.2):** A stronger vision encoder contributes disproportionately to fine-grained classification, with more limited effects on general VLM benchmarks when the model is properly pretrained.
- **Pretraining stage (§4.3):** Large-scale image-caption pretraining contributes substantially to fine-grained classification performance, while its effect on general VLM benchmarks is smaller.
- **Weight updating method (§4.3):** Updating both the connector and the LLM during pretraining yields stronger fine-grained classification than training the connector alone.
- **Data quality (§4.3):** In our ablations, changing the quality of pretraining and instruction-tuning data has a limited effect on fine-grained classification, especially when the LLM is frozen during pretraining.

Taken together, these contributions reposition fine-grained classification as an explicit VLM evaluation dimension and identify which parts of the VLM pipeline most affect it. The resulting picture is actionable: improving fine-grained visual knowledge requires attention to vision encoders and pre-training strategy, whereas language-model improvements raise performance more uniformly. These findings support the development of more vision-centric VLMs for real-world long-tail tasks that demand precise recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) connect visual inputs to language models, often using the LLaVA-style architecture (Liu et al., 2023) with three core components: a vision encoder, a language model, and an MLP connector between them. This design has enabled strong performance on multimodal tasks such as visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). The corresponding evaluation ecosystem has also expanded, with benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). Our contribution is positioned around what these evaluations make less visible: they primarily test reasoning and language understanding conditioned on images, while leaving core vision-centric capabilities such as object recognition and fine-grained classification under-measured. We address this gap by systematically evaluating and analyzing VLMs on fine-grained classification.

Fine-grained visual classification. Fine-grained classification is a long-standing computer vision task that distinguishes visually similar subcategories within broader categories, including bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). These benchmarks have been central to training and evaluating vision encoders such as ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Recent VLM evaluation protocols, however, have largely shifted attention from these datasets toward VQA and reasoning benchmarks. The practical importance of this omission is clear: domains such as medical diagnosis, food safety, and species identification depend on precise recognition under long-tailed distributions. Our evaluation therefore treats fine-grained classification not as an auxiliary diagnostic, but as a necessary lens for judging the vision-centric usability of VLMs.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies have begun to test VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), but they typically examine a limited set of models or datasets. We extend this line of work in two ways. First, we evaluate a broader collection of VLMs across multiple fine-grained classification benchmarks using a unified multiple-choice format. Second, we move from measurement to explanation by conducting controlled ablations that identify which model and training choices contribute to fine-grained performance.

A parallel literature analyzes the VLM design space, including architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). These studies primarily optimize or explain performance on general benchmarks that emphasize language, reasoning, or broad multimodal competence. Our work contributes a complementary analysis: we ask how base model selection, pretraining methods, and the composition of pretraining and instruction-tuning data affect fine-grained classification specifically. This framing helps isolate which design choices matter most for vision-centric VLM capability.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section makes our first contribution explicit: we evaluate whether the general progress of recent VLMs also appears in fine-grained classification. We motivate this evaluation, define the benchmarks and testing protocol, and then show that fine-grained classification exposes capabilities that current VQA-style benchmarks miss.

3.1 MOTIVATION

Fine-grained classification is a core computer vision task: models must distinguish visually similar subcategories within a broader category, such as bird species or car models (Wei et al., 2021). Bench-

marks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore served as standard testbeds for training and evaluating vision encoders. With large-scale supervised and self-supervised learning, encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have reached near- or superhuman performance on several challenging fine-grained classification tasks.

Our evaluation contribution begins from the observation that these same benchmarks are rarely central in VLM evaluation. Recent VLMs such as LLaVA (Liu et al., 2023) are usually tested on visual question answering benchmarks, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or on document and text-centric tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks are valuable, but they emphasize reasoning and language processing over fine-grained object categories. As a result, they can leave basic but important vision-centric recognition ability largely unmeasured.

Fine-grained recognition is also practically important for VLM deployment. Real-world inputs often follow long-tailed distributions in which visually similar categories imply different actions or risks. If a blind user asks whether a mushroom is safe to eat, the model must first identify the species before applying textual knowledge about toxicity; confusing a deadly destroying angel with an edible white button mushroom could be fatal. Similar stakes arise in medical diagnosis, where related diseases may require different treatment, and in driving, where confusing a “stop” sign with a “do not enter” sign could cause unsafe navigation. These examples motivate our focus on fine-grained classification as a safety- and reliability-relevant VLM capability.

3.2 BENCHMARKS

We evaluate fine-grained visual knowledge with four established object-recognition benchmarks that cover different recognition domains: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014). This benchmark set supports our contribution by testing fine-grained recognition across broad object categories, flower species, pet breeds, and food categories rather than relying on a single dataset.

Because these benchmarks contain 37 to 1,000 classes, directly presenting every class as a VLM multiple-choice option would be impractical and unlike common VLM evaluation formats. We therefore follow Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negative labels by image-label cosine similarity, and we combine those negatives with the ground-truth label. We use OpenCLIP to avoid biasing the negative selection against VLMs based on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), which are trained on different data. This protocol lets us evaluate VLMs in a native multiple-choice format while preserving much of the original classification difficulty: CLIP ViT-L/14-336px (Radford et al., 2021), for example, obtains 75.2% accuracy on the 102-class Flowers-102 benchmark and 78.3% on our 5-way version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use the model’s default prompt format from VLMEvalKit (Duan et al., 2024) and score accuracy by exact match against the multiple-choice options. As a vision-encoder reference point, we evaluate CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) with standard zero-shot classification: image and class-label prompts (“A photo of a [class]”) are encoded by CLIP’s image and text encoders, and the top-1 prediction is selected by cosine similarity.

To test whether existing VLM benchmarks already capture fine-grained recognition, we compare each model’s fine-grained average with its average score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024). This comparison is central to our evaluation contribution because it asks whether fine-grained classification is redundant with, or distinct from, the dominant benchmark signal.

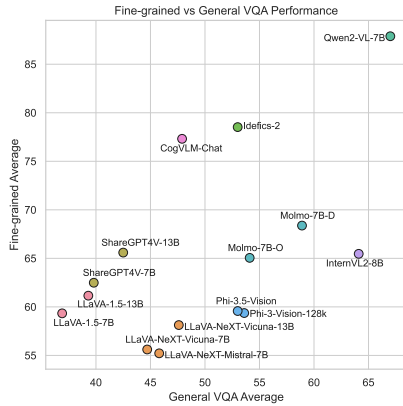


Figure 2: **Fine-grained classification exposes a capability not captured by general VQA averages.** Across recent VLMs, similar general VQA performance can correspond to very different fine-grained classification accuracy.

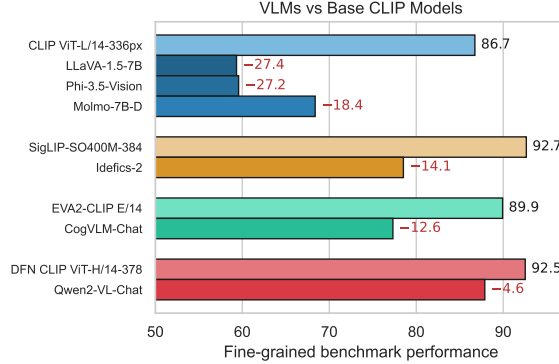


Figure 3: **VLMs often underuse the fine-grained knowledge of their vision encoders.** Qwen2-VL-Chat nearly matches its DFN-CLIP vision encoder, but most tested VLMs remain well below their corresponding CLIP baselines.

3.4 RESULTS

Figure 2 summarizes the first result: fine-grained classification does not track general VQA performance closely enough to be treated as already measured. The figure plots average accuracy across our four fine-grained benchmarks against average performance across eight existing VQA benchmarks for all tested VLMs. Models with similar general VQA scores can differ substantially on fine-grained classification. For example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both achieve nearly 48% average accuracy on general VQA benchmarks, yet their fine-grained classification accuracies differ by 19pp (percentage points): 77.3% for CogVLM-Chat and 58.1% for LLaVA-NeXT-Vicuna-13B. This gap supports our claim that fine-grained classification is a distinct dimension of VLM capability.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Figure 3 sharpens the second result by comparing VLMs with their own pretrained vision encoders. Because current VLMs are built on existing vision encoders, our multiple-choice format enables a direct comparison to CLIP-style zero-shot classification on the same tasks. The comparison reveals a substantial remaining gap. Molmo (Deitke et al., 2024), a recent model with strong VQA performance, reaches 68.4% average accuracy in our multiple-choice setting, while its CLIP ViT-L/14 (Radford et al., 2021) vision encoder reaches 86.7%. Even Qwen2-VL-Chat (Wang et al., 2024b), the best model we test, remains 4.6 points below its DFN-CLIP (Fang et al., 2023) vision encoder. This gap motivates the ablation study in the next section: if VLMs do not consistently preserve encoder-level fine-grained knowledge, we need to identify which design choices help recover it.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the evaluation in Section 3, this section contributes an explanatory analysis of why VLMs differ in fine-grained classification. We isolate model and training factors that could account for the observed performance gaps, focusing on base model selection and training methodology

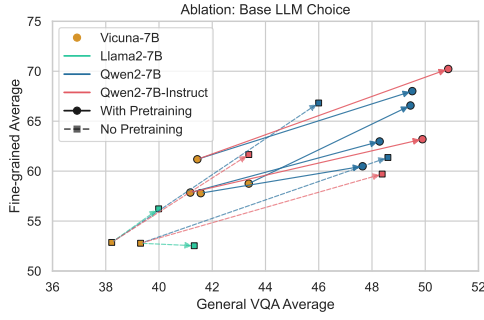


Figure 4: **Contribution of the base LLM.** Replacing Vicuna-7B in LLaVA with stronger LLMs improves both evaluation axes: switching from Vicuna to Qwen2-7B increases fine-grained performance by +7.5pp and general VQA performance by +7.7pp on average.

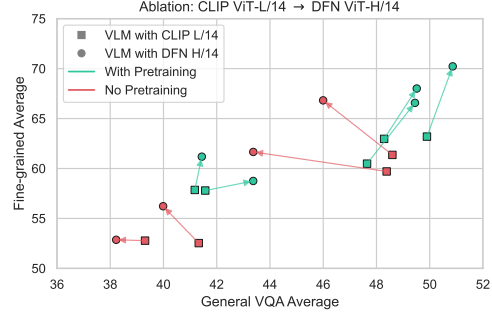


Figure 5: **Contribution of the base vision encoder.** Replacing CLIP ViT-L/14 in LLaVA with DFN-CLIP ViT-H/14 yields a more fine-grained-specific gain: with sufficient pretraining, fine-grained performance improves by +4.5pp while general performance improves by +1.2pp.

while keeping the overall LLaVA-1.5 alignment framework fixed (Liu et al., 2023). Unless otherwise noted, each model is trained for one epoch of pretraining when applicable, followed by one epoch of finetuning. We use LLaVA-1.5’s default hyperparameters, including learning rate and batch size, across all experiments.

For the fine-grained target of the paper, we report the average score across the four classification benchmarks introduced in Section 3.2. To distinguish fine-grained-specific effects from broad VLM improvements, we also report the average score on three widely used multiple-choice VQA benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks provide domain coverage and are widely used in VLM evaluation, making them useful references for determining whether a design choice improves fine-grained classification disproportionately or simply improves all scores.

4.2 BASE MODELS

Our first ablation group asks whether base model choices explain fine-grained recognition gaps. The observational results in Section 3 show that recent VLMs often pair newer LLMs with older or widely reused vision encoders, especially CLIP ViT-L/14 from Radford et al. (2021). We therefore separate the contribution of the language model from the contribution of the vision encoder.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as the controlled baseline, we replace only the LLM component while varying vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) used in LLaVA with Qwen2-7B (Yang et al., 2024), which is used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024). We also include Llama-2-7B (Touvron et al., 2023), the base model for Vicuna, and Qwen2-7B-Instruct, the instruction-tuned version of Qwen2-7B. This setup tests whether better LLMs specifically improve fine-grained recognition or instead act as a general VLM performance contributor.

Findings. Figure 4 shows that replacing Vicuna-7B with Qwen2-7B improves performance across configurations, yielding average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. Qwen2-7B-Instruct produces similar gains (+7.5pp and +8.1pp). Llama2-7B, the non-instruction-tuned base of Vicuna, provides more modest improvements, though the downstream implications of using a model without instruction tuning remain unclear. Importantly, even the lower-performing Vicuna-based VLMs consistently produce multiple-choice responses in the correct format. The contribution of the LLM therefore appears to be improved knowledge or option evaluation rather than improved adherence to the answer format.

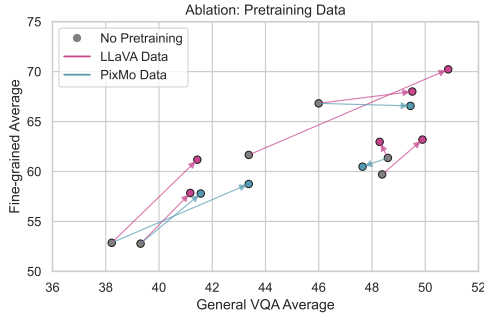


Figure 6: Contribution of pretraining data. Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) pretraining data produces similar gains. LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp, while PixMo data increases them by +2.4pp and +2.5pp.

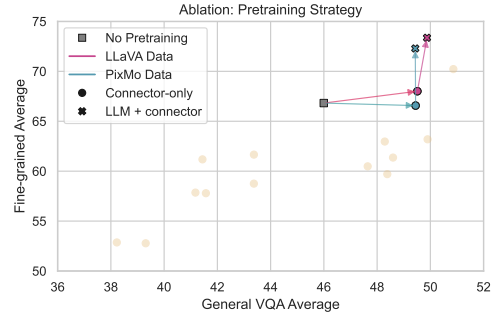


Figure 7: Contribution of the weight-updating strategy during pretraining. Compared with connector-only pretraining, unfreezing the LLM during pretraining improves fine-grained performance by +5.5pp without reducing general VQA scores.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. We next isolate the vision encoder, the component most directly tied to fine-grained visual information. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021). Qwen2-VL (Wang et al., 2024b), however, uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which has stronger zero-shot object-recognition performance on tasks such as ImageNet. We compare these two encoders across LLM choices and pretraining configurations.

Findings. Figure 5 shows that the vision encoder contributes most when it is properly integrated before finetuning. Without pretraining, replacing CLIP L/14 with DFN-CLIP H/14 reduces general VQA performance (-2.5pp) while modestly increasing fine-grained benchmark scores (+2.8pp). After connector pretraining, general VQA performance remains stable or slightly improves (+1.2pp), and fine-grained performance increases more substantially (+4.5pp). This pattern makes the vision encoder a fine-grained-salient contributor, but only when alignment pretraining gives the VLM access to the stronger visual representation.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Our second ablation group asks which training stages contribute to fine-grained visual knowledge. Recent VLMs are typically trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). Pretraining uses large-scale image-caption data to align the vision encoder and LLM through a captioning objective. Instruction tuning then uses curated multi-round, multimodal instruction-completion data to align the model with human-oriented tasks. We analyze both stages to determine whether fine-grained recognition is mainly a property of the base models, the pretraining alignment stage, or instruction tuning.

4.3.1 PRETRAINING

Setup. The pretraining ablations are designed around three contribution questions:

Table 1: **Summary of the main ablation contributions and comparison to existing models (bolded).** The table separates factors that broadly improve VLM scores from those that matter especially for fine-grained recognition: stronger LLMs help both axes, while stronger vision encoders and pretraining that updates the connector and LLM provide clearer fine-grained gains. Instruction finetuning contributes comparatively less.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

- Is pretraining still necessary for LLaVA-architecture models when both base models are already pretrained on billion-scale data?
- Does connector-only pretraining preserve fine-grained information as well as pretraining that also updates the LLM?
- Does higher-quality pretraining data improve downstream fine-grained recognition in this setting?

We test three pretraining-data configurations: LLaVA pretraining data (Liu et al., 2023), a CC-3M subset with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), PixMo-Cap with high-quality human-annotated detailed captions; and no pretraining. Most experiments follow Liu et al. (2023) and pretrain only the connector. We also test a strategy similar to Deitke et al. (2024): pretrain the connector for 20% of the steps, then unfreeze the LLM for the remaining 80%. Every configuration is followed by one epoch of finetuning on LLaVA finetuning data.

Findings. Figure 6 shows that connector-only pretraining generally improves VLM performance, with the exception of the Qwen2 and CLIP ViT-L/14 combination. The contribution is especially large for Vicuna-based VLMs, where pretraining improves fine-grained classification by +6.1pp on average and general VQA performance by +3.1pp. This result identifies pretraining as a major driver of fine-grained capability rather than a negligible alignment step.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

The data-quality comparison is more muted. PixMo captions are higher quality and more detailed than LLaVA’s web-scraped captions, but replacing LLaVA data with PixMo data changes fine-grained performance by -1.6pp and general VQA scores by +0.4pp. In our controlled setup, therefore, caption quality alone does not emerge as the main explanation for fine-grained performance differences.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We next test whether connector-only pretraining prevents the model from using higher-quality captions or richer visual-language alignment. To do so, we pretrain both the LLM and connector after a 20% connector-only warmup. Figure 7 shows that this training method improves fine-grained classification by +5.5pp without compromising general VQA scores. The gain appears with both LLaVA and PixMo pretraining data, suggesting that the decisive factor is updating the LLM during pretraining rather than avoiding lower-quality captions.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

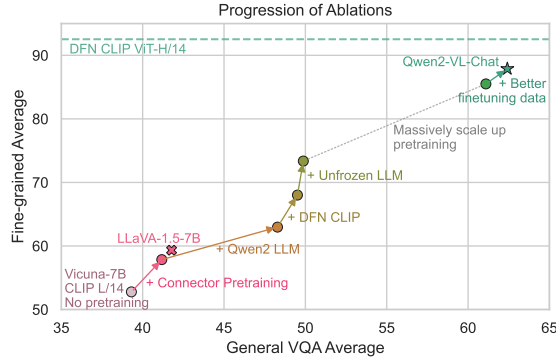


Figure 8: **Cumulative contribution of the ablated factors.** Starting from Vicuna-7B with CLIP ViT-L/14, successive changes to the base LLM, vision encoder, and training settings raise fine-grained classification accuracy from 52.8% to 73.4%, closing more than half of the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining gap primarily to Qwen2-VL’s much larger 1.4T-token pretraining scale.

4.3.2 FINETUNING

Setup. To isolate the contribution of instruction finetuning data, we start from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b), from which Qwen2-VL-7B-Chat is trained, and finetune it on LLaVA-Instruct data (Liu et al., 2023) using our earlier protocol. We compare this model with Qwen2-VL-7B-Chat, which uses 2M in-house instruction-finetuning examples (Wang et al., 2024b). This comparison asks whether the final instruction-tuning stage explains fine-grained performance once a strong pretrained VLM is already available.

Findings. Finetuning only on LLaVA data slightly reduces both evaluation axes. Fine-grained scores decrease by 2.4pp, from 87.9% to 85.5%, and general benchmark scores decrease by 1.3pp, from 62.4% to 61.1%. As shown in Figure 8, this is the smallest contribution among our ablations, indicating that instruction finetuning is less central to fine-grained knowledge acquisition than base model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 consolidate the ablation contribution of each factor. Starting from the LLaVA baseline architecture (Liu et al., 2023) with Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021), our controlled modifications raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The pattern across ablations is the main result of this section: switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately improves fine-grained classification, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for most of the general VQA improvement.

These gains do not fully explain the best observed VLM performance. Table 1 still shows a roughly 12-point fine-grained classification gap after our controlled modifications. We consider two plausible contributors to this remaining difference: architecture and pretraining data scale.

Although architecture may contribute, the evidence from our observational results (Section 3.4) and training ablations (Section 4.3) points most strongly to pretraining data scale. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) pretraining runs use fewer than 1M images and captions, corresponding to approximately 200M and 400M tokens. By contrast, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens. This orders-of-magnitude scale difference likely explains much of the residual gap and reinforces the section’s central contribution: fine-grained VLM performance depends not only on better components, but also on whether pretraining is large and deep enough to transfer visual knowledge into the multimodal model.

5 LIMITATIONS & CONCLUSION

Our analysis clarifies several contributors to fine-grained VLM performance, but it also has practical limits. The most important limitation is scale: because of computational constraints, our controlled training comparisons use fewer than 1M data points rather than the billion-scale or trillion-token pretraining regimes used by some newer VLMs. This means our ablations identify which factors matter in a controlled setting, but they cannot fully determine how the same factors behave under much larger pretraining budgets. In addition, newer work proposes alternative pretraining strategies that may interact differently with fine-grained visual understanding. These limits leave open future work on whether the contribution patterns we identify persist at larger scale and under different training recipes.

In conclusion, this paper contributes a more explicit account of fine-grained classification as an under-measured dimension of VLM capability. We first evaluate state-of-the-art VLMs on fine-grained classification benchmarks and show that these results are not well predicted by common general VQA benchmarks. We then use controlled ablations to identify which design choices are most associated with stronger fine-grained performance: stronger LLMs improve both evaluation axes, stronger vision encoders and pretraining choices matter especially for fine-grained recognition, and instruction finetuning contributes comparatively less. These findings make fine-grained visual knowledge easier to track in VLM evaluation and provide practical guidance for building more vision-centric models for real-world tasks that require precise recognition.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed ablation results across both evaluation axes.** This table reports the full set of controlled model and training variants used to support the paper’s contribution analysis.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed ablation results on general VQA benchmarks.** These scores provide the general-capability reference used to distinguish broad VLM improvements from fine-grained-specific gains.

A FULL RESULTS

This appendix supports the paper’s ablation contribution by reporting the complete numerical results behind the summarized trends in the main text. Table 2, Table 3, and Table 4 provide aggregated performance, general VQA benchmark results, and fine-grained classification results, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Detailed ablation results on fine-grained classification benchmarks.** These scores provide the vision-centric evidence for which model and training choices most improve fine-grained recognition.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

This appendix details the evaluation protocol that supports our main claim that fine-grained classification is a distinct and under-measured VLM capability. As summarized in Section 3.2, we evaluate VLMs and base vision encoders on four traditional classification benchmarks.

ImageNet-1K (Deng et al., 2009): ImageNet-1K is a foundational image classification dataset organized around the WordNet hierarchy. It covers a broad range of supercategories, many of which include multiple fine-grained subcategories. The ImageNet test set contains 50,000 images across 1,000 classes.

The ImageNet/WordNet hierarchy makes some classes too visually distinct from the rest of the label space to form hard multiple-choice questions; for example, only three classes appear under the “person” category. Because of this issue, and because the full dataset is large, we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Starting from 19,056 annotated examples, we reduce the pool to 5,457 images spanning 928 of the original 1,000 classes. Since this curation produces an imbalanced class distribution, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): Oxford Flowers-102 contributes a species-level recognition test with high intra-class variation and inter-class similarity. The dataset contains 6,149 test examples across 102 flower species.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): Oxford-IIIT Pet-37 contributes a breed-level recognition test with challenging variation in pose, lighting, and occlusion. The Pets dataset contains 3,669 test examples across 37 pet categories.

Food-101 (Bossard et al., 2014): Food-101 contributes a fine-grained food recognition test with substantial visual diversity within each class. The dataset contains 25,250 test examples across 101 food categories.

Question generation. For all fine-grained classification test sets except ImageNet, which is curated as described above, we generate hard negatives by running zero-shot classification with an Open-

CLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the top four incorrect predictions and shuffle them with the correct label to create the answer choices. This protocol preserves the contribution of the benchmark as a fine-grained recognition test while adapting it to the multiple-choice format used by VLMs.

Prompt formatting. In the multiple-choice VQA format, the question is always “What type of object is in this photo?” The answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because different VLMs are trained with different multiple-choice prompts and answer-format instructions, we use each model’s corresponding VLMEvalKit format (Duan et al., 2024), such as “Answer with the option’s letter from the given choices directly.”

For CLIP models, the comparison uses the standard zero-shot evaluation procedure rather than a question format. Class names passed to the text encoder are formatted with the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the initial observational comparison in Section 3, we use the OpenVLM Leaderboard numbers reported by VLMEvalKit (Duan et al., 2024). The default leaderboard average covers eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). We use this average only in Figure 2, where it serves as the general-benchmark axis for testing whether fine-grained classification is already captured by existing evaluations.

For all subsequent evaluations, including existing models and the models trained in Section 4, we use a narrower set of multiple-choice VLM benchmarks to make the comparison with our fine-grained multiple-choice protocol more direct. Specifically, we evaluate on three benchmarks.

MMBench (Liu et al., 2024b): MMBench contains 2,948 examples that test a wide range of VLM capabilities across perception and reasoning. Most examples involve coarse-grained perception or visual reasoning, although some test “fine-grained perception” in domains such as action recognition, attribute recognition, and OCR rather than the fine-grained classification datasets studied here.

MMMU (Val) (Yue et al., 2024): MMMU is a large-scale multimodal benchmark spanning many domains and applications. Although its test set contains 10,500 examples, most work reports performance on the validation split of 900 questions. MMMU evaluates general knowledge grounded in visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): MMStar contains 1,500 human-selected questions designed to cover diverse domains and low-level model capabilities. It provides a compact general VQA reference point for comparing against the fine-grained classification axis.

Evaluation procedure. For the general VQA benchmarks, we use the VLMEvalKit evaluation code directly (Duan et al., 2024). Existing models use their predefined evaluation procedures. For our trained models, we apply the LLaVA evaluation code because all of our models are finetuned on LLaVA-Instruct data.

C TRAINING

Our training details support the ablation contribution by keeping the framework controlled, so that changes in fine-grained classification can be attributed to specific model or training choices. We build on the LLaVA-1.5 training codebase (Liu et al., 2023), and most experiments retain the LLaVA architecture and training procedure.

Pretraining. Pretraining consists of one epoch on either the LLaVA pretraining data, which uses CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we use only the “caption” field and not the raw audio transcript. During pretraining, we use a batch size of 256. In most pretraining runs, we tune only the randomly initialized MLP connector with a learning rate of 1e-3. When ablating pretraining strategy, we first tune only the connector for 20% of steps as a warmup, then unfreeze the LLM and tune it as well with a learning rate of 2e-5 for the remaining 80% of steps.

Finetuning. Finetuning always uses one epoch on the LLaVA-1.5 finetuning data, which combines LLaVA-Instruct with VQA training data from multiple datasets to support benchmarking. During finetuning, the LLM learning rate is $2e-5$ and the batch size is 128. The vision encoder remains frozen during both training phases, ensuring that our ablations measure how existing visual representations are integrated into the VLM rather than how the vision encoder itself changes.

For the Qwen2-VL architecture ablation, we start with Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM. This setup isolates the contribution of finetuning data while keeping the pretrained Qwen2-VL backbone fixed.

Compute. All models are trained with 4 nodes of 4 A100 (40GB) GPUs each. Training typically finishes within 20 hours, with runtime varying by CLIP model size.

LLaVA vs. PixMo data. The main text compares LLaVA pretraining data with PixMo pretraining data to test whether caption quality explains fine-grained performance. LLaVA pretraining data comes from CC-3M and consists of web-scraped captions, which are commonly low in quality and relevance. PixMo-Cap is collected from human annotators who record a 60–90 second audio description of an image, which is then converted into an image caption (Deitke et al., 2024). A surface analysis shows that LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”